Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/11/2006
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Somatosensory Suppression and Prevention of Post-flight Re-entry Disturbances		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	93-OLMSA-01
Start Date:	01/31/2001	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul (Brandeis University)		
Grant/Contract No.:	NAG9-1263		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Our goal is to develop procedures for alleviating post-flight disturbances of posture and locomotion experienced by astronauts. We have shown that non-mechanically supportive contact of the hand with a stationary surface stabilizes postural and whole body movement control in 1) a normal 1g environment, 2) a rotating environment, and 3) parabolic flight. We have recently developed a hand-held sensory gyroscope that stabilizes posture in 1g. These observations provide the basis for four specific aims whose outcome will provide critical information for designing apparatus and procedures for suppressing re-entry disturbances after spaceflight and hastening sensory-motor readaptation to 1g. Aim 1: Measure how non-supportive contact with a surface and how a hand-held sensory gyroscope stabilize posture during exposure to different gravitoinertial force levels in parabolic flight maneuvers and in 1 g after landing. Hypothesis. We predict that both types of sensory cueing will decrease mean sway amplitude and total power of head and center of mass motion during in-flight force changes and will suppress aftereffects. Aim 2: Evaluate the influence of non-supportive hand contact and of a sensory gyroscope on deep knee bends in the 1.8g phase of parabolic flight. Hypothesis. We		

Task Description:	predict that both types of haptic input will make the intended vertical movements more regular, suppress unintended horizontal body displacement, and attenuate sensory-motor aftereffects. Aim 3: Determine how finger contact at non-supportive force levels and a sensory gyroscope attenuate body sway in a rotating environment. Hypothesis. We predict decreases in mean sway amplitude and spectral power of head and center of mass before, during, and after rotation relative to control levels without haptic contact. Aim 4: Measure the influence of non-supportive finger contact and of a hand-held sensory gyroscope on treadmill locomotion in a rotating environment. Hypothesis. Transitions to and from a rotating room will disrupt gait, increase center of mass excursions, and deviate gaze. We predict both forms of haptic cueing will suppress the primary effects of rotation and the aftereffects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on control of posture and movement in space flight environments has benefits for terrestrial applications. A deeper basic understanding of the role of somatosensory cues from the hands and feet holds great promise for developing mobility aids for people with balance disorders. Our research results are being used by others interested in basic mechanisms of human spatial orientation and motor control, clinicians, developers and practitioners of virtual reality, and space flight human factors professionals.
Task Progress:	No report received this reporting period.
Bibliography Type:	Description: (Last Updated: 03/31/2017)